CS 210

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Program 3 Analysis

1. Briefly define a *queue* data structure.
   A queue is a first-in-first-out data structure, meaning items are taken out of the queue in the same order that they're put in.

   What evidence did you find that a *queue* container was operating in the load balancer?
   The IDs, while not exactly, generally increased over the course of the output, showing that the older requests are being processed first, which fits the definition of a queue.

2. Briefly define a *stack* data structure.
   A stack is a first-in-last-out data structure, meaning the item most recently put into the stack will be the first one taken out.

   What evidence did you find that a *stack* container was operating in the load balancer?
   The IDs, compared to the queue, went up in large chunks of values batch-by-batch, rather than steadily increasing. This shows that the load balancer was processing the requests that came in most recently, which fits the definition of a stack.

3. Briefly define a *priority queue* data structure.
   A priority queue assigns data stored within a 'priority' value. Items are removed in order of priority

   What evidence did you find that a *Max priority queue based on process time* was operating in the load balancer?
   The requests being processed by the Max priority queue are values with higher processing times, showing that it is prioritizing the higher processing time.

   What evidence did you find that a *Min priority queue based on process time* was operating in the load balancer?
   The requests being processed by the Min priority queue are values with lower processing times, showing that it is prioritizing the lower processing time.

   What evidence did you find that a *Max priority queue based on process ID* was operating in the load balancer?
   The IDs of the requests steadily increase as the load balancer goes on, showing that it is prioritizing the lower IDs.

Evidence can be a description of the output specific to the data structure and/or an annotated image.